WORKED SOLUTION

STEP III 2001, Question 14

Probability and Statistics • Unbiased Variance Estimator

STEP III 2001, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q14

The key idea

Everything here turns on one fact: the "variance of the sample" uses divisor $n = 2$, so it underestimates the population variance, and doubling it puts that right. For two values the sample mean is the midpoint, so each value sits half the gap away from it. That makes V a function of the single difference $X_1 - X_2$, and the final probability becomes an area in the square $[0, a]^2$.

Solution

The variance of X

The density of X is $f(x) = \frac{1}{a}$ on $[0, a]$. By symmetry $\mathrm{E}(X) = \frac{1}{2}a$, and

$$\mathrm{E}(X^2) = \int_0^a \frac{x^2}{a}\,\mathrm{d}x = \frac{a^2}{3}.$$

So

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - (\mathrm{E}(X))^2 = \frac{a^2}{3} - \frac{a^2}{4} = \frac{a^2}{12},$$

as required.

The sample variance

For the sample X_1, X_2 the sample mean is $\bar{X} = \frac{1}{2}(X_1 + X_2)$, and the variance of the sample is the mean squared deviation from it:

$$V = \tfrac{1}{2}[(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2].$$

Now $X_1 - \bar{X} = \frac{1}{2}(X_1 - X_2)$ and $X_2 - \bar{X} = -\frac{1}{2}(X_1 - X_2)$, so both squared deviations equal $\frac{1}{4}(X_1 - X_2)^2$. Hence

$$V = \tfrac{1}{2} \cdot 2 \cdot \tfrac{1}{4}(X_1 - X_2)^2 = \tfrac{1}{4}(X_1 - X_2)^2,$$

which is the given form.

Ask yourself How do we find $\mathrm{E}((X_1 - X_2)^2)$ without a double integral?

Since X_1 and X_2 have the same distribution, $\mathrm{E}(X_1 - X_2) = 0$, so $\mathrm{E}((X_1 - X_2)^2) = \mathrm{Var}(X_1 - X_2)$. Because X_1 and X_2 are independent, their variances add (the minus sign doesn't matter, as $\mathrm{Var}(-X_2) = \mathrm{Var}(X_2)$):

$$\mathrm{Var}(X_1 - X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) = \frac{a^2}{12} + \frac{a^2}{12} = \frac{a^2}{6}.$$

Therefore

$$\mathrm{E}(2V) = \tfrac{1}{2}\,\mathrm{E}((X_1 - X_2)^2) = \tfrac{1}{2} \cdot \frac{a^2}{6} = \frac{a^2}{12} = \mathrm{Var}(X).$$

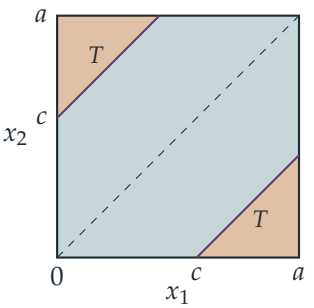
The expected value of $2V$ equals the quantity it estimates, whatever the value of a, so $2V$ is an unbiased estimator of the variance of X. (By contrast $\mathrm{E}(V) = \frac{a^2}{24}$, only half the true variance.)

The probability

Since $V \geq 0$ and $\frac{1}{12}a^2 > 0$, we may take square roots:

$$V < \frac{a^2}{12} \iff (X_1 - X_2)^2 < \frac{a^2}{3} \iff |X_1 - X_2| < \frac{a}{\sqrt{3}}.$$

The pair (X_1, X_2) is uniformly distributed over the square $[0, a]^2$, because X_1 and X_2 are independent and each is uniform. So the probability is the fraction of the square's area lying in the band $|x_1 - x_2| < c$, where $c = a/\sqrt{3} < a$.



The complement consists of the two shaded corner triangles T, each right-angled and isosceles with legs $a - c$. Their total area is $2 \cdot \frac{1}{2}(a - c)^2 = (a - c)^2$, so

$$\mathrm{P}\left(V < \frac{a^2}{12}\right) = 1 - \frac{(a-c)^2}{a^2} = 1 - \left(1 - \frac{1}{\sqrt{3}}\right)^2 = 1 - \left(\frac{4}{3} - \frac{2}{\sqrt{3}}\right) = \frac{2}{\sqrt{3}} - \frac{1}{3}.$$

That is, the exact probability is

$$\mathrm{P}\left(V < \frac{a^2}{12}\right) = \frac{2\sqrt{3} - 1}{3}.$$

With $\sqrt{3} \approx 1.732$ this is $\frac{2.464}{3} \approx 0.821$, so to one significant figure the probability is 0.8.

Checks

- The answer doesn't depend on a, as it shouldn't: rescaling X by a factor scales both V and $\frac{1}{12}a^2$ by its square.
- A simulation of 4×10^5 pairs with $a = 3$ gave a mean of $2V$ of 0.751 (target 0.75) and a proportion 0.821 with $V < \frac{1}{12}a^2$, against $\frac{1}{3}(2\sqrt{3} - 1) = 0.8214$.
- Sanity: the probability lies in $(0, 1)$, and replacing $\frac{1}{12}a^2$ by $\frac{1}{4}a^2$ (the largest possible V) makes $c = a$ and the formula $1 - (1 - c/a)^2$ give 1.

Common slips

- Using divisor $n - 1$ for the "variance of the sample": that gives $\frac{1}{2}(X_1 - X_2)^2$, not the stated $\frac{1}{4}(X_1 - X_2)^2$. The question wants the divisor n.
- Forgetting the absolute value, and so counting only one of the two triangles (which gives a probability below $\frac{1}{2}$).
- Writing $\mathrm{Var}(X_1 - X_2) = \mathrm{Var}(X_1) - \mathrm{Var}(X_2) = 0$.

Beyond STEP

The result here is the case $n = 2$ of a general theorem: for any sample of size n from a distribution with variance σ^2, the mean of $\sum(X_i - \bar{X})^2$ is $(n - 1)\sigma^2$. That is why the divisor $n - 1$ (Bessel's correction) gives an unbiased estimator. First-year statistics goes on to ask which unbiased estimator has the smallest variance. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both cover expectation and variance of sums in the style used here.

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